Viewing in the positive direction of the $y-$axis, divide the region into elementary vertical strips (thin rectangles) of height y and width Δx. Then, A is the limit sum of the areas of the vertical strips. Hence, we get $A = \lim \sum_{a\le x\le b} -y\Delta x = -\int_a^b y\,dx = \left|\int_a^b y\,dx\right|$.

Case (ii)

Let $y = f(x),\ a \le x \le b$ be the equation of the portion of the continuous curve that lies below the $x-$axis (that is, the portion lies either in the third quadrant or in the fourth quadrant). Then, $y \le 0$ for every point of the portion of the curve. It is important to note that y does not change its sign in the region. Consider the region bounded by the curve, $x-$axis, the ordinates $x = a$ and $x = b$. See Fig.9.9. Then, the area A of the region is found as follows:

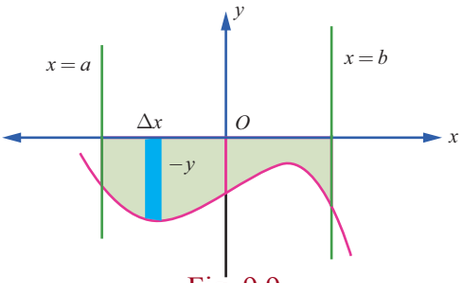


Fig. 9.9

Viewing in the negative direction of the $y-$axis, divide the region into elementary vertical strips (thin rectangles) of height $|y| = -y$ and width Δx. Then, A is the limit of the sum of the areas of the vertical strips. Hence, we get $A = \lim \sum_{a\le x\le b} -y\Delta x = -\int_a^b y\,dx = \left|\int_a^b y\,dx\right|$.

Case (iii)

Let $y = f(x),\ a \le x \le b$ be the equation of the portion of the continuous curve that lies above as well as below the $x-$axis (that is, the portion may lie in all quadrants). Draw the graph of $y = f(x)$ in the $XY-$ plane. The graph lies alternately above and below the $x-$axis and it is intercepted between the ordinates $x = a$ and $x = b$. Divide the interval $[a,b]$ into subintervals $[a,c_1]$, $[c_1,c_2]$, $\cdots$, $[c_k,b]$ such that $f(x)$ has the same sign on each of subintervals. Applying cases (i) and (ii), we can obtain individually, the geometrical areas of the regions corresponding to the subintervals.

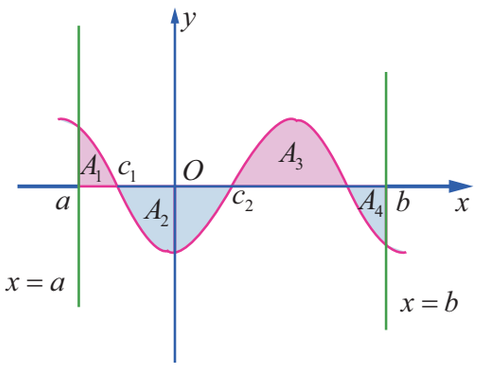


Fig. 9.10

Hence the geometrical area of the region bounded by the graph of $y = f(x)$, the x-axis, the lines $x = a$ and $x = b$ is given by $\left|\int_a^{c_1} f(x)\,dx\right| + \left|\int_{c_1}^{c_2} f(x)\,dx\right| + \cdots + \left|\int_{c_k}^{b} f(x)\,dx\right|$.

For instance, consider the shaded region in Fig. 9.10. Here A_1, A_2, A_3, and A_4 denote geometric areas of the individual parts. Then, the total area is given by

$$A = A_1 + A_2 + A_3 + A_4 = \int_a^{c_1} f(x)\,dx + \left|\int_{c_1}^{c_2} f(x)\,dx\right| + \int_{c_2}^{c_3} f(x)\,dx + \left|\int_{c_3}^{b} f(x)\,dx\right|.$$

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